THE WORLD, THE FLESH AND THE METAL*

THE PREROGATIVES OF SYSTEMS

By STAFFORD BEER
Managing Director, SIGMA (Science in General Management, Ltd.), Croydon, Surrey

The Seminal Idea

COMMANDING concepts arise in science which seem to roll up the universe into a ball, declaring: it works thus. I am thinking of such notions as inertia and entropy. The names of these concepts do not denote any discoverable thing out there in the world at all. They denote sophisticated ideas that scientists have about the way the world is arranged, or perhaps read into the world. What is out there is a manifestation of the idea; and that manifestation is pervasive, once the idea is sufficiently clear to be called obsessive.

The old way of talking about these concepts was to legislate about them, and to set up Laws of Nature. Who passes these laws is not clear, but certainly God alone may repeal them. So clustering about the concept of inertia are the laws of motion—together with a considerable body of amendments which encompass relativity; and clustering about the concept of entropy are the laws of thermodynamics—together with special versions of the enactments for other territories, such as statistical mechanics and information theory. These laws seem to be very unusual ones indeed; for there is no way of repealing them—once the Deity is defined in their own terms as a store of negentropy.

Now concepts of this kind are very difficult to understand, because there is no ostensive definition. Hunting down inertia is like hunting down the snark; the only one who actually saw it "softly and suddenly vanished away". The result is it takes a long time before the concepts which are most potent in the scientist's interpretation of the universe are manipulated with ease in our thinking. As evidence of this contention, consider the adjectival form of the terms already mentioned. Physics acquired the concept of inertia in 1687, through the *Principia*. But it was a hundred and sixty-two years before anyone invented the adjective, and spoke of 'a weight of inertial resistance'. Where would science be without that word to-day, a mere hundred years later? As to the word entropy, the centenary of the birth of which falls in 1965, we are still so unsure of ourselves that there is no official record that the adjective has been coined yet. I have in fact coined it myself, and offered such phrases as 'entropic drift'; but no one seems to want them.

The point of all this is to suggest that the concept of system is another such seminal notion, which we are only now beginning to understand. This word has also been used since the seventeenth century; and is certainly well understood in the rather special sense of an assemblage which is usually small, usually connected together deterministically, and usually well defined. But we cannot roll up the universe into that sort of ball, nor declare it works thus. Patently the universe is not, nor is it compounded of, that sort of system. Yet a system in another sense it clearly is, and any part of the universe is clearly a system in that same sense. Secondly, the syntactical clue to our unease in the manipulation of this wider concept is again present. For the adjective which means 'pertaining to a system' is not 'systematic' (which means something quite different) but systemic; and that is a word one seldom hears. It took a hundred and eighty-four years to achieve the adjectival form this time, even in a limited physiological context. As an adjective of general relevance, the word systemic has been with us for only a hundred years, and its use is now marked 'rare' in the dictionary.

In short, we do not think about the systemic nature of things, we prefer to consider them in isolation. Stepping smartly down from these philological clouds, I point to a number of practical examples. The brain, a company's profitability, a computer and the National Health Service are all integral systems. But we talk gaily about the motor cortex as if the rest of the brain were irrelevant to movement. We talk about the cost of making a product as if it did not matter what opportunities to make other products were foregone in making this one. We talk about the reliability of an electronic component as if this meant something independently of the failure rates of associated components in an operational ensemble. We talk about the cost of prescriptions under National Health as if this were independent of the doctors' attitude to whether the patient bears a charge or not.

The seminal notion of system-as-pervasive in science is really very new, and we are not trained to think in systemic terms because our scientific approach has been analytical to the ultimate degree. We took things apart, historically, and described the atomic bits. We did our experiments, historically, on these bits—deliberately holding invariant the behaviour of other bits with which in fact they were systemically interacting in real life. Now advances in statistical method have enabled us to be more ingenious than this to-day; we can afford to let a number of things change at the same time, and sort out the meaning of the experiment by multivariate analysis. But this is a mastery of technique, not a change in methodology. The result is that our only approach to system is through aggregating bits. No wonder that we do not advance in understanding really big systems, then, and that the very connotation of this word system includes such question-begging epithets as small, determined and well defined.

The concept of system in its widest meaning is a cybernetic concept. Big systems are the topic of cybernetics. The systemic character of the world is the clue to its control; just as the inertial character of the world is the clue to its movements, and the entropic character of the world is the clue to its energies. Thus the contention is that the 'laws of control' are the prerogatives of system—wherever and whenever it is defined. And this is why I have taken liberties with the catechism in the title. For the prerogatives of system are pervasive, whether we speak of the world at large, or the flesh of our own bodies, or the metal of the ironmongery we invent as engineers.

Methodology of Models

If this concept of system is so new and so important, how shall we attack it, how comprehend and manipulate it? In a paper written five years ago[1] I gave a philosophical approach to this problem. To-day the approach will be essentially mathematical.

* Substance of the 1964 Stephenson Lecture delivered before the Stephenson Engineering Society, University of Newcastle upon Tyne, on November 26.

When an engineer looks at a big system (as already defined here) and recognizes its systemic character, he may well perceive its structure in terms of servomechanics. He draws rings around various activities, or sub-sets of events, in his mind, and observes that they are linked together by messages. That is to say, that the input to one sub-set is the output of one or more other sub-sets. For example, production is an activity which responds to a demand expressed by a market, while selling is an activity responding to a production output. The engineer may then recognize that what happens in each box of the system may be regarded as a transfer function applied to the signals that connect it to other boxes. If the macrosystem he is thinking about can once be defined as closed, as in the case of the marketing-production-marketing example, he is in business as a model-builder. The process soon becomes exciting. In order to attain to an ever-more-accurate account of what the big system is doing, the model-builder may insert all sorts of boxes in his design, until the model is very complicated indeed. Very soon he will see the need for local regulatory devices in the microsystems which comprise the macrosystem. This will lead to the insertion of closed-loop feed-back circuits, in which outputs are monitored by comparators, and inputs modified by some function of the error signal thereby formed. It is often possible to obtain a strikingly good analogue, expressed in servo-mechanical terms, of a fairly big system of world events in this fashion.

In fact, this is one example of the way operational research works. The scientific model conceived by an operational research team will depend on the kind of system being studied and on the versatility of the team itself. We are all conditioned by our training and experience. It can scarcely have been fortuitous that Prof. M. J. Lighthill chose to represent the flow of traffic with a model from fluid dynamics, or that Prof. A. Tustin sought to model the economy by just the kind of servo-system that I have been describing. This is not because the big systems of real life are 'really' hydraulic or electrical in character, but because Lighthill and Tustin are the distinguished men they are, and because the systemic character of the world is pervasive. This is why an operational research group is supposed to be inter-disciplinary. It is my job to hold back a colleague who is determined to model a management structure by an anthropological model taken from the tribes of the Western Pacific, if this turns out to be rather unhelpful. Equally I expect my colleagues to intervene if my own habit of modelling company organizations on neuro-physiological systems runs away with me. They always do intervene, as it happens; but I still think that the brain, supported as it is by many million years of research and development, must offer good advice on how to make a control system viable. Which just shows how preconditioned I am. Anyone who wishes to examine more closely how operational research makes use of models from science may refer to another paper[3], in which various examples are given—including the electrical circuit of a servo-model of an actual production system.

But, assuming that this talk of models is now understood in a general sense, I propose to examine rather more rigorously what is going on when a model of one activity is taken from another activity. Here it should be borne in mind that our comprehension of anything at all is likely to be expressed to ourselves in terms of something else which we know that we understand; which is another way of saying that everything is a model of something else. If you should ask: what constitutes the prior structure of which other things are models?—then maybe the answer belongs to psychoanalysis, and must be expressed in terms of elemental and traumatic experiences.

Consider a big system out there in the world. It may be a firm interacting with its environment of markets in a search for survival; it may be a company competing with another company for a larger share of a zero-sum pay-off; it may be a capitalist economy in which a socialist chancellor sits like a Maxwell demon increasing the entropy of wealth. (Please note in passing how, simply in order to nominate three samples of big systems, I have already invoked three models: from ecology, from game theory, and from thermodynamics.) In trying to understand these things as systems, we must detect in them whatever is systemic. This sounds tautologous, but it is not really. For we do not have full knowledge of what these systems are like—they are immensely complicated; and we do not have full knowledge of what constitutes system anyway—science is not yet omniscient, and we ourselves are relatively ignorant.

Now I shall discuss this problem in the language of group theory, for reasons which will be made specific later on. The point is that we need as rigorous a statement of this complicated methodology of models as we can get. It is in fact high time that some such attempt were made, for the very word 'model' has fallen into an unhelpfully casual usage. Some people use the word as synonymous with 'mathematical equation'; others treat the word as if it meant 'hypothesis'. A model is neither of these things. It is something that opens up a virtually new way of doing science.

Thus if we call the set M of elements a the totality of world events which we propose to examine, then the systemic configuration of events which we know about is a sub-set A of set M. If we call the set N of elements b the totality of systemic science, then the configuration of system which we ourselves understand is a sub-set B of set N. The process of creating a systemic model may then be described as a mapping f of A into B. By this I mean that for every element $a \varepsilon A \subset M$ there exists a corresponding element $b \varepsilon B \subset N$, and thus $b = f(a)$. The image of the sub-set A, namely, $f(A) \subset N$, is the model. If we are able to exhaust the elements of A and to nominate their images in B, we have every hope of creating an isomorphic model. This means that there exists a complete inverse image of B under mapping f in M, so that $f(A) \subset N = f^{-1}(B) \subset M$. This is the state of affairs, expressed group-theoretically, which the operational research man is trying to reach.

Now an isomorphism is important because it preserves the structure of the original group in the mapping. Typically, if it is possible to perform additions inside set M, those additions will remain valid when the same operations are performed on the images of their elements in set N. It is this persistence of relationship when the mapping is done which makes a model operate as a model. So, if a_1 and a_2 when added together equal a_n in set M, it can be shown that $f(a_1)$ plus $f(a_2)$ must equal $f(a_n)$ in set N. Now comes the interesting comment. The conditions can be set up in which the same answer $f(a_n)$ in set N is obtained from the mapping f whether the transformation is effected before or after the mapping occurs. That is to say, we may either add the original elements in M and transform the answer under f, or we may transform the original elements first and then add them. The result will be the same. Formally: $f(a_1 + a_2) = f(a_1) + f(a_2)$. When one group is mapped into another group and this condition is generally fulfilled, the mapping is called homomorphic.

These elementary definitions are included so that the argument can be made quite clear. Because it is possible to coalesce elements of M before transforming them, without losing the capability of a mapping to preserve structural relationships as discussed, it is clear that a homomorphism may have fewer elements than its inverse image. In the case of the model, then, the mapping of A into B turns out to be a mapping *onto* a sub-group of B. Isomorphism turns out to be a special case of homomorphism, in that $f(A) \subset B$ turns out to mean $f(A) = B$: the one–one correspondence of elements with which we begin is maintained. But for any other sub-group of B

other than B itself, homomorphism involves a many-one correspondence, and the inverse mapping $f^{-1}(B)$ will not exhaust the elements of A.

It is suggested, then, that the models of big systems that we entertain are homomorphisms of those systemic characteristics of the big system that we can identify. The homomorphic group $f(A) \subset B \subset N$ is the particular model we use. It is in practice extremely difficult to include in this model all the features recognized in A, and typically we do make the many-one reductions mentioned. Thus, for example, we undertake production costings as if the behaviour of all three shifts in a works were indistinguishable, and as if two similar products were identical, and as if materials were consistently uniform—although we actually know that none of these simplifications is true. Then the effectiveness of the model as predictive depends on the choice of an effective transformation by which to map. If we add up the outputs of three shifts and then transform the answer by some mapping into the model, it is no use supposing that any calculation, comparison or prediction undertaken in the model can be worked backwards through an inverse mapping which will distinguish between the shifts. On the other hand, it is necessary to handle only a third of the elements we know about inside the model. A definite choice has been made to jettison modelling-power in favour of economy in the recording and handling of data. This is acceptable, so long as the choice is deliberate rather than accidental, and so long as it is remembered as a limitation in the model.

Secondly, however, there is a further loss of modelling power in the facts that A is a sub-set of M and B is a sub-set of N. Now an interdisciplinary team of scientists can minimize the losses of modelling power due to $B < N$. Because such a team can examine all the major sub-sets of N before deciding to use one specific group B; it may even experiment with other groups too. But the losses due to $A < M$ are more serious, and may be disastrous to the exercise. For if what we recognize in a big system is not what is really important about its systemic character, the ability to predict A may not help much in M. In other words, A is itself a homomorphic mapping of M, and one which by definition we cannot properly specify. Remember that M–A was acknowledged to be systemically unrecognized from the start. We may know that our knowledge of a big system does not exhaust it, without having the faintest idea of the character of the knowledge that is missing.

It is hoped that this attempt somewhat rigorously to formulate what goes on in model-building will prove helpful in pin-pointing what we can and cannot do. The ordinary operational research exercise works, and we can see why. It is possible to advance what we understand about a stock-holding system, for example, to the point where A approaches M asymptotically. It is possible to examine most B of N, which is to say most scientific approaches to the scientific totality of understanding about such systems. If we know what the stockholding system has to do, if (as the operational research man would say) we can define its criteria of success or objective function, then we can define a homomorphic mapping f of $A \simeq M$ onto $B \simeq N$ which preserves the stochastic relationships in which we are interested. More especially, we can do this in such a way that the inverse image of B under mapping f yields a set $f^{-1}(B)$ of elements in the real system M which are useful.

The difficulties about doing successful operational research in various circumstances can now be made quite specific. First, the modelling will not on the average work well if N–B is large: this happens if the operational research team is not corporately versatile. Secondly, the modelling will not work at all unless f is well defined: this entails good empirical research into what the system really has to do. Thirdly, the predictions of the model will be of no actual use if a modelled outcome $\varphi(b_1 \cdots b_n)$ turns out to have a pragmatically undiscriminating inverse $f^{-1}(\varphi)$ in A. This also entails good empirical research into the forms of many-one reduction. Fourthly, the modelled predictions though useful will not exert what could be called control unless the $M \rightarrow A$ homomorphism captures the systemic character of the big system *in extenso*. This again appears to be a matter for good empirical research, although there is more to say.

Contrary to increasingly current belief, then, operational research is empirical science above all. The mathematical models dreamed up in back rooms are useless unless they can meet the four kinds of difficulty enumerated, and this cannot be done remotely from the world. But that is by the way.

Basis of Systemic Control

If the ordinary operational research study works for the reasons just given, it is easy to see why we encounter difficulty in extending its scope to the really big system. It is not the difficulty of procuring the homomorphic mapping $A \rightarrow B$, which practice and scientific skill can encompass. It is the difficulty of procuring the homomorphic mapping $M \rightarrow A$. Now if we rightly called the mapping $f(A) \subset B \subset N$ the model, it is the complete inverse image of this model in M that constitutes a sentient control. The point is one of some subtlety, and repays a little thought. It says that deliberative control can be exerted only through the systemic connectivity of the big system that we recognize; and, moreover, that we are enabled to exert that control only via our model of how it works, with an efficiency that depends on the relevance of the mappings f and f^{-1}.

Interestingly, this deliberative control is not primarily what does control any big system. It is not adequate to the task. No one imagines that the set of actions open to being taken by a Chancellor of the Exchequer constitutes the control of the economy; it constitutes at best an error-controlled feed-back loop of either positive or negative sign. And the managing director who believes that he constitutes the control of the firm is deluded for the same reason. The reason why this must be so is given by the volume of information defining the behaviour of a big system. In cybernetics, we call the number of distinguishable states that a system may take up its variety. Clearly, the big system proliferates variety; and clearly this variety is not readily handled by a control system which cannot absorb it—which is choked by it, or confused by it, or otherwise overwhelmed. I have already said that ideally A should tend to equivalence with M under some homomorphic transformation. This was meant to ensure that the systemic structure of the big system would be preserved in the understood version of its nature which was to be modelled. What we now go on to say is that the measure of variety evinced by the big system must in some sense be paralleled in whatever is actually controlling it.

Examine this proposition in terms of the thesis already developed. We want a model of the big system M competent to determine its actual rather than deliberative control mechanisms. Since we do not, by definition, know what these are, we need to preserve a one-one correspondence of elements. What then is isomorphic to a big system ? It is inconceivable that we should go away and construct another system of equivalent variety—for this really is a big system, and our model would have to be as large. The group-theoretic formulation gives the clue to the answer. Cayley's theorem declares that every finite group is isomorphic to a certain group of permutations of itself; and it is part of the proof to show that one of these is the identical permutation. In other words, the isomorphism we seek for the big system is itself. Hence the big system M, to which is clamped another system A which is in fact itself, is to be called controlled.

Do not imagine that this conclusion is vacuous. It is, after all, the realization that an isosceles triangle ABC is congruent with the identical triangle CBA that enables us to prove, without construction, that the base angles are equal. If we try to give material substance to the mathematical abstraction of the big system's isomorphism with itself, we shall, so to speak, have to think of cleaving the system in half through its plane. Then every element is twinned, and the idea of what constitutes effective control suddenly becomes blindingly clear. Every player is marked by another player; or, to use another metaphor, half the population are policemen keeping one-to-one guard on the other half.

The law of control which has emerged is, I believe, a new formulation of Ashby's law of requisite variety[3]. It says, in effect, that control can be exercised only by a controller having at least as high variety as the system to be controlled. This is what happens in big systems, because it must. Two halves of the system are monitoring each other; though woe betide any glib observer who imagines he can determine the boundary between them. It is not as easy as that. The two sub-sets are interpenetrated; and, one suspects, elements move in and out of both sub-sets continually. Requisite variety in a controller is the power to absorb the variety proliferated by what is to be controlled, but this power is disseminated throughout the system rather than being concentrated in a control box.

However, there is better news—if we are cautious. Electrons are continually being exchanged between any object and its surroundings, but this does not in practice deter us from identifying objects and declaring that they persist. If we had sub-atomic magnitude ourselves, it would matter very much: doubtless we should not be able to identify these objects at all. If, then, we look at our big system as compounded of sub-systems which exhaust the whole, we may well make arbitrary rulings about what constitutes one part and what another, but we shall have preserved requisite variety. For although there is no means of saying exactly what is included in any sub-system, and although we may be quite unable to analyse the relationships which subsist between the elements of these sub-systems, we shall still be able to talk about control. Control is precisely the stable state of the variety interactions between the sub-systems nominated.

How does this account differ from the original one in which the engineer looked for the servo-system within the big system? In the first place, it is exhaustive—by logical definition. But, more important, it does not see communication as a signal passing from one transfer function to the next in a thin white line. It sees communication as interpenetrative between the sub-systems which are richly interconnected—(in the limit) element by element. Yet, more important still, it sees systemic stability itself as the object of the system, and not the holding steady of an arbitrarily assigned output—which is the usual criterion of control engineering.

These outcomes of the discussion give most of the leads which are required to answer the initial question: how do we attack, comprehend and manipulate the systemic character of the universe? They involve, I fear, a revolution in thinking for the trained scientist. No longer do we aggregate bits of the system, which have been adequately investigated, into even larger well-understood systems, which eventually (but when, indeed?) ought to approximate in size and scope to the big system under investigation. Instead, we begin by carving up the not-understood big system into interacting sub-systems, and we ask behavioural questions about the resulting homoeostats—which are the stably interacting parts. This can be done, and can result in useful statements about the big system itself. Comprehension begins with observations about the way in which two halves, or better still n $n$ths, of the big system interact, although there is virtually no knowledge about the contents or internal structure of those halves or $n$ths. Comprehension is extended by resolving (in the optical sense) these structures to an ever-increasing degree. But we should not imagine that we shall ever reach the atomic parts and their relationships—which are the very entities with the close examination of which orthodox science is accustomed to begin.

Anyone who finds this too puzzling should forget that he is a scientist for the moment and remember that he is a man. How is it that we can make useful predictions about the way each other's brains work? We do a reasonable job; for people do not constantly surprise us, rather the dispiriting reverse. Perhaps we do it by drawing inferences from other people's behaviour, their responses to our stimuli, over a period; perhaps we argue by analogy with our own responses; perhaps we do both. In any event we do it by making models, by mapping behavioural structures on to each other; by juxtaposing; by interacting. What we quite clearly do not do is get down to that mess of pottage, a brain newly excavated from its cranium, with a microscope. If we did, and escaped hanging, we should still not learn enough to make predictions about people's behaviour. The classical method of enquiry would encumber us with the investigation of the basic nerve cell, the atomic component, called the neurone—from which we would work outwards to the big system, the cerebrum. According to the American cybernetician McCulloch (who very possibly knows) the transfer function of a neurone is to be defined only in terms of an eighth-order non-linear differential equation. There is no guarantee that any two neurones have an identical transfer function, and the brain contains ten thousand million of them. Next, we should need to know how they arranged and interconnected. The fact that all our lives we lose about a hundred thousand a day (they just pack up) is another complication. No: it just will not do.

Return then to the notion that requisite variety is supplied in natural systems, and that deliberative control is based on a model of a sub-set of the actual control mechanisms. What the would-be controller, or manager, has to do is to intervene in these natural processes rather than to impose some control arrangement on them. Good managers intuit that this is their task. The trouble is that so many of the 'experts' who seek to advise them do not always understand this point, unless the big system is a person. In that special case, human prerogatives are invoked to say that people ought not to be controlled by the imposition of restraints on their freedom: I speak of an ethical imperative. Certainly, some restraints are imposed, in the interests of the community at large; but if we want a particular man to do a particular act we do not contemplate using a pistol, or drugs or hypnosis. We have to organize the system of which the man is a part in such a way that he does what we want of his own accord. This is the right way to control any big system, such as a firm or an economy; not because of ethical imperatives but because of cybernetic laws. For the implicit control of the big system, which requisite variety guarantees, is carrying the system to a more stable condition naturally.

The management task is to arrange its pay-off objectives in relation to the system's own built-in objective of stability. The best way to express this rule is to say that the system has an informational entropy which carries it towards a more probable state. Management needs to arrange the system so that the most probable state is the desired state. This means that the natural force of entropy is harnessed as a control force in the system. It also means that the system offers us an amplifier of our control signals free of charge, and a complete set of appropriate control circuits for distributing control signals to the right places in the system—even though we have no knowledge as to where the right places are.

If this sounds too good to be true, remember that the job can be done effectively only if we use a model which is an appropriate homomorphism. The 'experts' of whom I complained just now are people who imagine that their heads contain this model, and that their own *savoir-faire* is a sufficiently good account of the inverse mapping. Why do economists disagree as to the correct governmental action in a given economic crisis? Surely it is because they pass off hypotheses about the way the country works as models with known mappings; so that any two of these non-models will read different deliberative control policies into the inverse image, which are quite likely to be inconsistent if not contradictory. The same is true of advisers to firms, transportation experts and so forth. The control of big systems demands a prior investigation of the natural control homomorph, and a scientific account of what entropies are involved. From this we can learn how to use the entropic drift as a control device. It is the only way to generate requisite variety in deliberative control for the inverse mapping $A \rightarrow M$.

Systemic Control in 'The World'

When we come to consider applications of these arguments to the world in general it is natural to turn to national affairs. For here are the big systems with which it is most important that we learn how to deal. There are, of course, international affairs, if we wish to set the sights higher.

Now it is possible to argue that most of the troubles we encounter in deciding on policies at the national and international level derive from our habit of dividing the system, which is too big and too complicated to handle integrally by classical methods, and trying to compute with the pieces—or sub-systems. The reason why a world which is affluent in some places and penurious in others cannot find a method for controlling food supplies is surely thus. There is no need for a third of the world to be hungry, for a baby to die of starvation every forty-two seconds. But the boundaries drawn around the sub-systems, around each country and around political blocs, are too rigid. In terms of the entropies which system-oriented scientists perforce discuss, these sub-systems operate within virtually adiabatic shells. Hence there is no chance of levelling out food supplies, which constitute the energy of the total system, or (by a mapping) of levelling out human happiness and human misery. Nor is there any obvious political technique by which a Stephenson Lecturer can propose to start engineering with these conventional packages. In national affairs, there is. Let us consider some examples.

It is by now a platitude to say that the plan conceived for the future of the railways is probably wrong, but that if it is wrong the fault lies with a Minister and not with the Rail Board. This is said because the railway chiefs were given certain tasks concerned with railway profitability; they were not asked to take into account the integral system of national transportation. Here, I should agree, is a perfect example of what the logician calls *fallacia divisio*: separating components which ought to be kept together. Keeping those components together is indeed a systemic imperative. Where transport is concerned, we have most notably fallen foul of the classical mental discipline which tries to build large systems as aggregates of carefully studied elements. That people should in general be able to get from one place to another is important. But this aim has become subservient to bogus optima, whereby each element of the system must itself be 'economic'. On the railways and in the air-lanes, this criterion has been applied. So far as roads are concerned, attempts are often made to cost the journeys of the private motorist, and to show that it would be cheaper for him to travel by public transport.

Beginning with this case, we see how the systemic law is ignored. If a man lives in circumstances which necessitate his owning a car, then the car is part of his transportation system. To cost a journey which he could have made by train on the basis that the capital cost of his car ought to be apportioned by mileage to this journey is a silly piece of accounting. He has this car anyway. The cost of his journey includes wear on the car perhaps, as well as running cost; but it certainly does not include a share of capital cost. On this basis, he will prefer to travel by road on economic grounds. Thus begins the destruction of that non-homomorphic model which calculates travel in terms of a fixed price per mile. The cost per mile of any kind of journey depends on the systemic context. It follows that the prices charged per mile by transport undertakings ought not to be invariant, as they basically are. For example, since the profitability of an air journey depends on the proportion of seats occupied, and since potential passengers are to some extent motivated by economic considerations, it follows that the price per seat for a given journey should vary with the bookings made. Thus the price paid for an air ticket reserved months in advance would include a component for the value to the passenger of a guaranteed seat. As the time of departure approaches, the price for a seat should fall. A passenger prepared to risk making no flight in exchange for a cheaper ticket would bring the plane's occupancy to an economic level. Note that a continuously computed fare based on these factors would at last begin to engineer with the uncertainty of demand, founding itself on high-variety systemic interactions between the air service and need, and between the air service and competition.

This kind of approach takes note of how the world is; it begins to construct a homomorphic model which can be used for deliberative control. At present, our controls are attempted impositions of rules which do not pay off because they struggle against the tide of entropy. In such cases the proper resort of management is to engineer with the structure of the system, until the entropies created in interaction with the world yield what the airline wants, namely, a profit. A similar argument may be applied to the railways, which have tried by advertising to condition the passenger to believe that 'it is quicker by rail'. Often, it is not quicker. Often, the potential passenger is less impressed by speed than by comfort, cleanliness and a sense of privacy. His selection entropies may then be drifting in a direction that does not conform to the systemic reference frame set up by the railway management. The battle for custom is then lost. If a passenger prefers to be fussed over by a pretty air hostess than insulted by a ticket collector, it is no good protesting that he should have more sense: and the best conclusion is not necessarily to close the railway line. It may be to abandon the attempt to match a high-variety demand with a low-variety service. Not even a monopoly can force through the repeal of the law of requisite variety.

Nor, indeed, can a whole powerful ministry repeal that law. The task of controlling road traffic, when attempted by policemen or by local regulators such as traffic lights, is a high-variety task unsuccessfully handled by low-variety control devices. It is quite obvious that requisite variety cannot be obtained to handle a problem of the magnitude now made manifest without massive sensory input to large-scale computing engines. Yet the use of computers is regarded, at worst, as a novel gimmick; and, at best, as an expensive innovation which perhaps must come as part of a technological upsurge. We never see the argument, based on fundamental thinking, that there are systemic imperatives which can be analyse dand must be met. It is not surprising, then, that we see no attempt to derive the homomorphic model of national transportation which would serve as the basis for well-informed research into the deliberative control of movement. Such a model must be global, following the principles adduced in the preceding section. Ideally, we should use satellites to photograph the ebb and flow of traffic, in order to obtain the systemic conspectus that we need. There are less ambitious ways of doing so. But the London Traffic Survey does not even have this aim. It represents the old-fashioned, non-systemic approach of trying to understand the elemental components of the system. Life is too short.

The result, moreover, is that we do not begin to approach the cybernetic problems of traffic control which really are important. If we had a homomorphic model of the interactions of sub-systems of traffic, we could begin to work on the serious issues that affect the dynamics of all big systems. (So far we have not discussed dynamics at all, only the structural aspects of systemic control.) Now the great problem in cybernetic dynamics is this. Every sub-system, however defined, is itself seeking a local stability. The big system as a whole seeks stability in terms of such local equilibria. But the homomorphic model of the big system, as expressed through the interaction of sub-systems, is difficult to express in homogeneous terms. The reason is timing. If a sub-system is to be regarded as having an input which is the output of another sub-system, then it is vital that the relaxation time be consonant with the periodicity of the perturbations arising from the input sub-system. Otherwise, the receiving sub-system will remain in perpetual uncontrolled oscillation. Moreover, from the point of view of deliberative control considered as an inverse mapping of the model, there is a problem of scale. One sub-system will see an interconnected sub-system as pumping variables into its equations of motion; another will see the same input as setting its basic parameters. Which is which will depend on the relative relaxation times of the receiving sub-system to the two input sub-systems. There is no need, surely, to press these problems to an audience of engineers. The point is that we have no idea how to handle them, because we have not even recognized that they exist, because we have not even begun to think out the model that reflects them.

They are, nevertheless, the problems which have to be solved: and if they exist for traffic management, they exist for economic management too. Moreover, if it is wrong to cut up the transportation system into separate units, it is wrong to cut up the national fuel system into separate units too. Who is there in the country at all who understands the systemic interaction between coal and gas and electricity and oil and atomic power? No one, so far as can be discovered, has begun to consider the homomorphic mapping of this integral whole. The examples are capable of indefinite extension. They seem to indicate a wholesale failure to understand anything at all about the prerogatives of systems. I hope it will now be appreciated why such care was taken to expound a methodology which can at least begin to discuss such matters, and why it is important to see that there has to be a reversal of the classical trend whereby science advances from the element to the system. It is because we have always done this, I repeat, that we have not to this day encompassed the really big system at all.

This difference of approach may be examined in the context of another big system which is particularly important to us all, namely, technical education. The future of this system and the demand on it were investigated in a famous White Paper, it may be recalled. The method of this enquiry, it will be no surprise to recollect, was to build up the 1970 requirement for technologists from the atomic components of employment as it existed in the past. There was no attempt to assess systemic interaction, nor the systemic trajectories of the future. In another paper[4] I have tried to expose the fault in this enquiry, and to show how the cybernetician would approach the task of modelling the actual control mechanisms involved.

There is one last point to make about the prerogatives of systems in the world at large, and it is this: Because of the habit of dividing, for administrative convenience, what should be kept together, we often identify as special to a part what is really a function of the whole. This causes us to try to measure, as a sub-systemic variable, some factor which might best be understood as a behavioural (that is, output) characteristic of the entire big system. An example which springs to mind from recent work concerns marketing. Everyone who spends money on sales promotion would like a measure of advertising effectiveness. This is, inevitably, sought in the change in sales value that follows in the wake of a campaign. But it is notoriously difficult to measure this change and to relate it to the appropriation—because of systemic interaction. There are time-lags, competitive effects, changes in conditions, and so on. It has become increasingly likely, to me, that the desired measure is not only impossible to make but actually invalid. The flavour of the argument, which cannot be set out here at length, is obtained by asking where in the engine of a car is to be found the vehicle's speed.

This sort of consideration can be contemplated with competence from the vantage point of an effective homomorphic model alone. For example, mention has already been made of the National Health Service. This centres on a system of patients and doctors interacting, each set providing the other with requisite variety. But there are several sub-systems interlinked with this one, and interpenetrating with it element by element. Two most notable such sub-systems are the regional hospital organizations and the pharmaceutical profession. All these sub-systems seem often to be uncoupled in the administrative and indeed the professional mind. Then the interesting question is how to resolve such a problem as the appropriate payment of all concerned. Here perhaps is another example of a global behavioural characteristic, masquerading as an entire set of sub-systemic variables. Certainly it is a problem for engineering in large systems to resolve, rather than a matter of local accountancy.

Systemic Control in 'The Flesh'

There are various levels at which we can examine the relevance of systemic laws to animate systems, considered as distinct from the world that is not aware. If we chose to discuss the individual living creature, we should find many examples to enlarge our cybernetic understanding. For *Amoeba proteus* and *Homo sapiens* have this in common: they are both very big systems compounded of sub-systems, and their character is systemic or it is nothing. Indeed, if we nominate viability as the objective of any organism, whether animate or pseudo-animate (as a firm or an economy), it is natural to turn to biology for advice. Cybernetics has sprung from biology, and has quickly involved physics and engineering, rather than the reverse, for perfectly sensible reasons. It is wise not to forget this, nor to ignore its lessons.

But the example of the operation of systemic laws 'in the flesh' that I shall take here comes from a different level of viable organization. I want to talk about the process of decision in a management group. Now decision theory, so-called, we have; it is an investigation appropriate to mathematical statistics. But decision theory assumes a uni-dimensional model of the world; it says that we know the terms in which we should rightly talk, and that we know the interactions that are important; it is not systemic. The process of decision in a management group is, on the contrary, systemic in character. The reason why it often takes many years to reach a complicated decision is because this is not understood.

Consider, then, such a decision. For example, where, and when, and how should the country add to its steel-making capacity by building a new plant? All manner of people will have views on this, arguing from every sort of expertise. The management problem is to find coherence in the spate of advice. Some of it will be factual; some of it will be inspirational masquerading as factual;

most of it will be contingently factual—that is, 'what I am saying is true, if something else is true, but not absolutely'. It is the contingent advice which specifies the system of decision; but unfortunately most of it is not recognized as contingent. It is here, then, that operational research is needed, so that the decision space and its configurations may be rightly understood. But operational research will not take this decision. It will be taken by a large number of people, each of whom is guaranteeing some sub-decision of the whole. The control problem for the highest authority is to monitor the decision-taking procedure to which all the sub-deciders are contributing.

The classical approach to this situation is to regard a very minor sub-decision as a basic component, and to build up a highly complex total decision from these elements. Hence some junior metallurgist somewhere may declare it obvious that a metallurgical reaction is best obtainable in a certain way, and begin to dictate the whole technology to be used. An economist looks at the siting problem in terms of the technology he knows about; and so on. But, according to the arguments in this article, we should not build up a very complicated specification from single components of decision. There is another way of looking at the matter. We do not really start with a set of elemental propositions, but with an infinitely large number of high-variety, integral alternatives. We do not really finish with a highly complicated specification for action, but with a unique integral answer. So the decision-taking procedure is not a process of building up more and more variety, but of eliminating more and more uncertainty. How can this procedure best be controlled ?

We said that the decision space had to be defined and that it is composed of possible answers. That is to say, every alternative integral answer might be considered as a small square. The decision space would then be the totality of these alternatives—say, a million small squares. For convenience, we might consider them as arranged in a square array, or matrix. This decision space will be ranged over by any decision procedure, in a hunt for the right answer. A management group normally undertakes the task as a heuristic search, which goes on by elimination. The average number of steps taken before the right answer is recognized is evidently half a million, since there is no reason why the right small square should be encountered either sooner or later than any other. True, sub-sets of small squares may be eliminated at once if they share a recognized characteristic regarded as damning; on the other hand, there may be a great deal of wasteful hunting back and forth between two or more sub-sets of small squares, each of which looks as if it might include the right answer. Note that this procedure is precisely obeying the control law of requisite variety. The variety of the decision space (a million alternatives) is matched by a control process organized to examine (if necessary) a million squares.

However: the natural variety proliferation of the world can be matched by variety generation—which I define as a deliberative control device. The decision space in our example is a square array. If we can identify the co-ordinates of this matrix as logical variables, then we can specify any small square (including the right answer) as a logical couple. To do so, we must search the co-ordinates. But the variety of each co-ordinate is 1,000; so the search space is 2,000 (instead of 1,000,000), and the expectation is that we find the answer in 1,000 steps instead of in 500,000. Thus the variety generator fulfils the law of requisite variety (since it has the selection power to discriminate between all the alternatives) but is a deliberative controller five hundred times as efficient as the natural controller.

All this is said to demonstrate the point. In practice, there is no reason why the decision space should be two-dimensional. Indeed, if a given decision had only two logical variables, it would not be a difficult decision. Then it has n logical variables, and the decision space is an n-dimensional space. Then the variety generator is yet more efficient. For the expectation that a set of integral alternative answers of high variety V will yield up the wanted answer by a general search is $\frac{1}{2}V$. Whereas the expectation for an n-dimensioned search with variety generators is $n/2.V^{1/n}$. We have found in principle a method of dealing with large-scale decisions which capitalizes on the systemic structure of the problem. It is just not true that decision-takers normally do take this advantage, by some analogue of this algorithm, as empirical research into actual decisions reveals. But it remains to show how we shall take advantage of it in practice.

First of all, it is necessary to identify the logical variables which are relevant. That is to say, we need to know all the dimensions of the decision: what sorts of things have to be specified to make the decision definite. This process is not one of listing all the statistical variables which affect the magnitudes eventually to be used: that side of the job is done by applied mathematics. We want to know what factors must be accounted for before the decision counts as a decision. The steelworks decision, for example, is not specified unless it includes the location of the works, its output of ingot tons, its range of products, and so on. These are the logical variables. The systemic interaction of these variables defines the configuration and dimensionality of the decision space. The statistical variables will eventually determine the optimal *modus operandi* of the steelworks decided on.

The number of logical variables is n, that is the dimensionality of the decision space. But the variety-generator notion enables us to observe that the n-space involved is not just a square, nor a cube, nor a hypercube: it has a configuration. Suppose, for example, that it is sub-decided for sufficiently powerful reasons to have a steelworks of given ingot tonnage, or at any rate within some range of tonnages. This automatically delimits a range of reasonable costs, a collection of possible sites, and so on. We do not yet know what the elements of these sub-groups are; but we can see that the decision space is inwardly contorted in various ways. There exist logical inferences, which we have not worked out, for any combination of logical values that may be fixed for any of the logical variables. In that case we must try to express the decision space in terms of its logical variables to give structure to the systemic configuration which the real world tends to proliferate.

The way to do this is through symbolic logic. This is the language of science for handling relatedness: conjunctions and disjunctions, implications and inclusions. The decision we are after has a known number of logical dimensions, for we have listed them. The systemic structure of the decision will be given by a logical formula stating how the logical variables are interrelated. Now this is a model in the sense defined. It requires a mapping of the way the external world works in relation to this decision. We have its dimensionality; we have its configuration. Next, we must measure its variety. In fact, this is not too difficult. Some logical variables are binary, in which event they offer a single alternative—a one-bit decision. For example, the steelworks must either be integrated with an ironworks or not. Other logical variables may propose a group of choices. For example, there is a finite number of steel-making processes between which, or between combinations of which, a decision must be taken. Suppose that, in a given case, there are eight choices. The variety of this decision is 8, not 2. This is a three-bit decision: it is composed of three binary decisions: $2^3 = 8$.

We have now discovered how to measure the variety of the decision, which is to say the uncertainty which has to be removed in searching for the completed specification. The measure is, of course, an entropy; not a thermodynamic entropy, nor the entropy met with in statistical mechanics, but an informational entropy—an entropy of selection. But, however we view it, the statistic that computes its value is the same. It is: $H = -\Sigma p_i \log_2 p_i$, where p_i is the probability of the ith choice. (We can ignore Bolzmann's constant in this application.) For every logical variable in the logical formula, we must compute this entropy.

In the case where the choice is a straight alternative between equiprobable outcomes, the entropy is $H = -(0{\cdot}5 \log_2 0{\cdot}5) - (0{\cdot}5 \log_2 0{\cdot}5) = 1$. This is the classic one-bit decision, and it shows why we are accustomed in cybernetics to use logarithms to the base 2: the answer is measured in the units of information theory, namely, bits. Now, in any application, there may be reasons why one alternative is more probable than another, and this will disbalance the equation. For example, if a committee of ten people is divided nine-to-one on the first showing of hands, the uncertainty to be resolved is less than one bit. If we wish, we may use this kind of fact as a measure of probability. Thus: $H = -(0{\cdot}1 \log_2 0{\cdot}1) - (0{\cdot}9 \log_2 0{\cdot}9) = 0{\cdot}469$ bits. But in general, we may consider that all alternative choices are nominated as equally probable. In the case that there are m alternatives, the equation is $H = -m.1/m \log_2 1/m = \log_2 m$, which is, of course, simple to compute.

By summing the selection entropies appropriate to each logical variable, we obtain a measure of the uncertainty that has to be eliminated in reaching the decision. This quantified formula is now a control device of requisite—and measured—variety. For any attempted specification of the decision, put forward by any party to it, has a certain entropy that can be computed; and the difference between this number and the number of the logical formula measures directly the inadequacy of the proposed answer. Moreover, if we start with a given level of uncertainty, we may progress our decision-taking by computing the uncertainty eliminated by any agreed sub-decision. Evidently, when all the uncertainty has been eliminated, the entropy is zero—because the probability that this is indeed the answer is unity. In other words, the decision has been taken.

In practice, the rate at which uncertainty is eliminated by a decision procedure will be some kind of negative exponential decay function. Empirical research will establish the family of curves which it is appropriate to use as control optima, and fiducial limits can be set up. Then we have an error-controlled feed-back loop with which to monitor the progress of the decision-taking. In a particular sphere of decision, moreover, particular clues to the uncontrolled situation will emerge. For example, whatever the decay function for uncertainty, it will surely be monotonic-decreasing. Hence if, in the course of plotting the progress of the decision-taking, a sudden upsurge in the graph should occur, it is clear that something has gone wrong. Uncertainty is being imported into the system. This will usually be due to an unacknowledged change in the objectives of the decision. It is clear evidence that the logical formula ought to be re-examined, and perhaps subdivided. Here again is one of those control devices which operate on unanalysed systemic blocs, rather than on elemental components.

This procedure, which is here made public for the first time, seems to be a powerful tool. It is simple to use, though difficult to understand for anyone who has not undertaken the preliminary methodological thinking given in this paper. Its particular merit, I believe, is its capability to quantify a logical structure. Hitherto, it has been virtually impossible to disclose a mode of inference which would couple together, in one technique, the power of logic to specify structure (which is much greater than that of mathematics) and the power of number to compute comparisons and the parameters of control (which is much greater than that of logic). It is a tool that has special value in the context of national decisions, such as those discussed in the preceding section; because the specification of each such decision is so great, and the parties to it so numerous and so dissociated organizationally.

Systemic Control in 'The Metal'

Having discussed, however cursorily, the relevance of the pervasive notion of system and its natural laws to the world and the flesh, we finally reach the metal. The metal stands for machinery, for devices, for artefacts built by man. Here, surely, we do understand the laws of system; here, one might hope, there is nothing to say.

Unfortunately, this is not the case. Where relatively small systems are concerned, the laws of system as understood by control engineering are happily ascendant. But we make big systems, by the criteria used before, in hardware too. In particular, we make computers, and we sponsor automation. It would be uncouth to repeat now all that I have so often said and written before on these topics[5]. But a few points ought to be made in the terms established by this text.

First, a system of automation is a deliberative control, that is, as has been stated, an inverse mapping of an M on to an A. It is vital that the laws of requisite variety be upheld; yet no automatic controller yet devised begins to match the variety of an ordinary works. Consequently, the Buddhist principle of control has generally been applied. It is the central, perhaps the only, idea of Buddhism that since we can never reach our targets of desire (and are thus condemned to frustration) we should alter those targets to equate with whatever we in fact attain. If what we are and enjoy becomes our target, behold, we have attained it after all. This is exceedingly good cybernetics; requisite variety is ensured, not by generating variety in our lives, but by eliminating variety in our ambitions. It seems to me that our notion of automation depends on this rule: the factory must behave in the way we have engineered to control it.

The result of this is that automation has been applied only to relatively straightforward, mass production situations—which are low-variety worlds. The jobbing works, on which the British economy largely depends, has so far been immune—*qua* big system. Sub-systems have been automated, it is true; and the computer-controlled machine tool is an example. But if we rightly understand and apply the systemic laws of cybernetics, it will not prove impossible to make automation much more general. The secret is, of course, to use computers as variety generators. Operational research does precisely this with its simulations; but engineering seems to think more in terms of variety reduction.

Thus we have the spectacle of the logical space of a general purpose computer programmed down to low-variety control, as, for example, in its application to the cut-up line of a rolling mill. This seems to be as wasteful of decision-taking capacity as is the programming down of a high-variety human being to being a machine-minder. There are no ethical objections in the former case as there are in the latter; but there are much stronger economic objections. These are the source of the marginal pay-off from computer applications in real time, which in my opinion is in turn a major cause of the slow advance of automation itself. The on-line control computer ought to be sensorily coupled to events in real time, and ought to have a motor output similarly coupled. But in between it ought to be uncoupled to a greater extent than is required to undertake a few simple-minded calculations on its inputs. In particular, it should have the facility to simulate the outcomes of possible decisions and to choose between them. This is not at all difficult to do; it is not even difficult in principle to equip the machine with a learning facility through which to profit from its own experience. The trouble is that managements do not commission such applications, and engineers do not offer to make them. The reason is, I strongly urge, not the imagined difficulty, but a very real insensitivity to the systemic rules of the game. As we are trained in the scientific methodology of analysis rather than synthesis, so we are trained in the Buddhist principle of control. We are variety murderers to a man.

An interesting example of this contention is found in the machine translation of language. A language is a high-variety system *par excellence*. The history of machine translation shows how science ignored this fact, and set out to translate automatically using low-variety control. It obtained, with chagrin but quite predictably, very bad translations. But a start had been made. Gradually the research began to acquire for the automatic translator some of the variety it had manifestly required from the outset. Crude syntactical paradigms were replaced by more elaborate ones; stored vocabularies grew. But the methodology is the old familiar one. We are aiming to reach requisite variety by the exhaustive enumeration of possible states, and we are still a long way off. Yet the computer as such is by no means short of variety; the inadequacy has been in the variety of the structuring of the decision space. We are not using variety generators, once again.

Surely the clue to this problem is the way in which people learn to translate (note the word 'learn'). We do not set out to programme children to be good translators. The trick is, rather, to give them a rudimentary grammar and vocabulary with which they are encouraged to experiment. They proceed to generate variety all right; the only trouble is that the output is unacceptable. So we monitor the output, using a skilled linguist (called teacher), who himself has requisite variety. He observes the defects in the homomorphic mapping of acceptable language on to the child's language. He then corrects the child by reinforcing successes with rewards and damping down failures with reproof. In systemic terms he uses what I have called the algedonic (pleasure/pain) loop. It is not difficult to devise an artefact of this loop inside a computer: all one needs is a simple conditional probability model to supply the criterion of retrieval. Instead of the rather naïve notion that a stored linguistic association is either present or absent, we want the stored material to be more or less likely to emerge—in accordance with a balance of systemically connected probabilities, gained by experience, that the answer will be praised or blamed.

I do not think we should be too diffident in seeking to imitate in such ways what are apparently human prerogatives, for they may be nothing more than systemic prerogatives—which is what this lecture is about. Before becoming emotional about words such as 'think' and 'create' it is well to define in behavioural terms what counts as thinking or creating. If this proves to be possible, it seems to me inevitable that science and engineering between them can devise a machine with comparable behavioural characteristics. This is all we need. It is interesting to ask whether such machines are 'really' thinking and creating, and if so how, and if so whether human beings do the job in the same way or some other. But to declare that machines can think, or that machines cannot think, seems as meaningless as it is pointless. I should refuse to mount the scaffold on behalf of any such proposition. But bear this in mind: whatever we may reckon as to genuinely human prerogatives, and as to the possibly divine nature of man, what we see men actually do is organized to be done by their brains: and the brain itself is a control machine.

A good and a final example of what I mean concerns the trick of procuring arbitrarily reliable outputs from a machine with arbitrarily unreliable components. Considered in engineering terms, the brain is a frighteningly defective apparatus, and yet it serves us well. It was because John von Neumann posed the question: "How do we retain any semblance of humanity when we have had too much to drink?" that cybernetics found out how the brain works this apparently impossible trick. As is now quite generally known, the answer is that it generates the requisite variety both structurally and operationally through redundancy: redundant nodes, redundant channels, redundant control complexes, redundant calculations. We have the theorems by which all this works; we even have a special logic of neural nets by which to describe it. But, again, engineers do not want to use all this knowledge. They have been brought up to believe that components ought to be reliable, and they design as if they were reliable. But no one has ever seen an actual component that did not have a measurable, finite probability of failure: and in a big system, a complex ensemble of components, these probabilities are multiplicative.

The result is that we refuse to send one redundantly controlled rocket to the Moon to make a few observations; we cannot afford to accelerate 'unnecessary' componentry to escape velocities. It is apparently better to fire six rockets, beautifully engineered and shorn of redundancy, none of which gets there. Again, we cannot afford the high cost of redundancy in a railway signalling system. After all, if perchance it should go wrong, the accident inspector will declare that it was 'one of those things'—a one-in-a-million chance. Personally, I regard myself as one of several million passengers; and my ghost would doubtless take a poor view of these odds. Yet again, then, we cannot bear to tell industrialists that they should consider installing redundant automation, because the expense might frighten them off the project. We prefer that they should refuse to accept automation on the grounds that it might go wrong.

It does not, in fact, seem to be totally ludicrous that we should be thinking in terms of a new kind of systems engineering, in the metal—as in the world and the flesh. It would be an engineering appropriate to big systems, aimed at providing requisite variety for control, equipped with variety generators, redundant circuitry, and an interpenetrative connectivity. Consider the following vision. Here is a vast plant, a huge, complex interacting system. Its engineers have built control equipment into it, of a subtle cybernetic kind. The plant is therefore self-regulatory (we know that trick). It is to some extent self-organizing (we begin to know this trick too). The plant is adaptive to environmental changes (we have discussed how this could be done through algedonics). But the built-in control systems keep going wrong. The teeth of the cogs drop out, the transistors fail, wires come adrift, whole packages burn out. The engineers, however, have seen all this happen in the brain. They have noted that brains do not continually 'go off the air' for maintenance: and so they have copied the brain's trick for handling unreliability, no longer complaining that things sometimes go wrong when manifestly they cannot always go right.

This works continues to work, despite all. It is not attended by an army of white-coated technicians armed with soldering irons, as some prophets have supposed. For the built-in control equipment is designed to outlast the plant. Do not say this is impossible; you and I are the evidence that it is possible. Do not say that it is impracticable; we have not even tried, and the principles are known. Say, if you must, that you are not going to do it, that you cannot afford it, or that it is time to go home; let us know where the onus lies.

There are systemic laws to apply, which could change the face of Britain—as George Stephenson once changed it. Twenty years before the *Rocket* ran that illustrious man nearly dug out the brain drain; for he was disheartened, and contemplated emigration. Some of us here have done the same. May we instead learn from his example, and see the visions through.

[1] Beer, Stafford, "Below the Twilight Arch: A Mythology of Systems"; chap. 1, *Systems Research and Design* (John Wiley, 1961).
[2] Beer, Stafford, *The Cost Accountant*, **42**, No. 6, June 1964 (also obtainable from SIGMA, Ltd., as a pamphlet).
[3] Ross Ashby, W., *Design for a Brain* (Chapman and Hall (London), 1952: revised edition, 1960).
[4] Beer, Stafford, *Operational Res. Quart.*, **13**, No. 2 (June 1962).
[5] Beer, Stafford, *The Manager*, October and November, 1963; *Metra Journal*, **3**, No. 1 (1964) (also obtainable from SIGMA, Ltd., as a reprint).